



COMPUTER & INFORMATION TECHNOLOGY — FULL MIND MAP

कंप्यूटर एवं सूचना प्रौद्योगिकी · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway



Topics in this chapter:

1. Computer Origins, Hardware, Input/Output & E-waste
2. Internet Basics, Networks, Wireless Technologies & VPN
3. Indian Supercomputers, Computer Types, Y2K & Printers
4. WWW, Protocols, Search Engines, Browsers & Network Models
5. Malware, Cybersecurity & Operating Systems
6. Binary Data, Memory Units, LAN, CPU & Hardware/Software
7. Applications, Computer Generations, IC Chips, Modems & Memory
8. Programming, Domains, Communication, AI & Digital Technologies
9. Devices, E-Governance, Multimedia, Video, E-mail & Mobile ICT
10. Mobile Platforms, Cybercrime, Data Security & Computer Networks
11. IT Companies, Social Media, E-Commerce & Cloud Computing
12. Web Acronyms, Logic Gates & Emerging Technologies

1. Computer Origins, Hardware, Input/Output & E-waste

- 1837 ke aasapaas Analytical Engine ki concept ki wajah se Charles Babbage ko widely father of computing maana jata hai. programmable mechanical computer ki concept ka invention karne ke liye Charles Babbage ko universally Father of computers maana jata hai. Charles Babbage ne Analytical Engine ko design kiya tha, jo modern digital computing ki foundation hai, jisse unhe Father of computers ki title mili. Second World War ke dauran Lorenz cipher ko todne ke liye British codebreakers ke liye Colossus, electronic digital computers ki ek series develop ki gayi thi. [Q1–Q4]
- Early electronic digital computer (jaise Colossus) Second World War ke dauran cryptanalysis purposes ke liye Britain me develop kiye gaye the. IBM System/360 1964 me announced ek highly successful mainframe computer family tha. Ise widely solid-state logic technology use karke banaye jane wala pehla major digital computer system maana jata hai jo modern IC-chip based architecture jaisa tha. motherboard computer ke andar central physical hub ki tarah kaam karta hai, jisme iske basic circuitry components (circuitry components) hote hain. pen drive, ya USB flash drive ek highly portable, removable secondary data-storage device ki tarah kaam karta hai. [Q5–Q8]
- Ek mouse ek essential hardware input device ki tarah kaam karta hai, jise originally 1960 ke decade me Douglas Engelbart ne banaya tha. mouse ko strictly input device ke roop me classified kiya gaya hai, jise computer ko command bhejne ke liye design kiya gaya hai. Listed options me se, monitor, printer aur plotter ko strictly output device ke roop me classified kiya gaya hai. plotter, printer aur monitor un hardware ke example hain jinka use computer-generated output ko display karne ya present karne ke liye kiya jata hai. [Q9–Q12]
- digital camera optical information capture karta hai aur ise system me feed karta hai, jo mainly input device ki tarah kaam karta hai. computer ko data paas karne ke liye mouse, keyboard aur Scanner specially input tools ke roop me kaam karte hain. keyboard sabse primary (quintessential) input device bana hua hai, jisse users computer system me text aur command sequence enter kar sakte hain. keyboard ke saath, mouse modern computing setup me sabse zyada use kiye jane wale input devices me se ek hai. [Q13–Q16]
- Dr. Douglas Engelbart ne 1964 me pehle primitive computer mouse prototype develop kiya tha. Double-click karne se computer ki Central Processing Unit (CPU) ko immediate electrical signal jata hai. Modern computer keyboard usually ek standard USB (Universal Serial Bus) Port ke through system unit se connect hote hain. computer systematically information ke storage aur analysis ka kaam karta hai. Encryption aur Password use karke, ye data ki secrecy/privacy maintain karta hai. Iske alawa, network se jude system harmful malware ya virus ke attacks ke prati sensitive hote hain. Sabhi statement computing features ka accurate description karte hain. [Q17–Q20]
- computer ghatakon me Beryllium, Cadmium, Chromium, Mercury aur Lead jaise toxic metals hoti hain, jo hazardous e-waste ka development karti hain. Heptaaklor ek chemical pesticide hai aur electronics se related nahi hai. Plutonium ek highly radioactive radioactive material hai jo nuclear use ke liye hai, consumer electronics me normally used nahi hai. Isliye, (1, 2, 3, 5, 6) sahi group hai. [Q21]



EXAM TRAP — Q11 — Hybrid Input/Output Device

- ▶ Touch screen display output bhi deta hai aur touch input bhi leta hai; isliye 'not an output device' wording source me disputed hai.

2. Internet Basics, Networks, Wireless Technologies & VPN

- Internet ko vaishvik level par ek huge 'network of networks' ke roop me samajha jata hai. Ye standardized Internet Protocol Suite (TCP/IP) use karke countless private, academic, commercial aur government systems ko connect karta hai. Internet ek huge, global digital infrastructure ki tarah kaam karta hai jo huge information resources ko seamless roop se share karne ki suvidha ke liye computer ko connect karta hai. Digital Subscriber Line consumers ko broadband Internet data transmit karne ke liye conventional copper ki Telephone laaion ke high voice-frequency band ka use karti hai. Historically, Dial-up Service (Dial-up service) web connection ke sabse dheeme roop ke roop me jaanee jati hai, jo lagbhag 56 Kbps ki maximum speed deti hai. Zyada modern faster options ke opposite, active hone par ye phone line par exclusive use kar lete thi. [Q22–Q25]
- Kisi web domain ki primary pravishti ya greeting screen ko 'Home page' kaha jata hai. Ye visitors ko saaita ke deeper, specific vargon tak le jane wali Instructionsikaa ki tarah kaam karta hai. Lagaataara, kisi core web Address ko hit karne

wale kisi bhi browser dwara present main early point 'Home page' (Home page) hota hai. Standard network mapping sanrachanaaon me level, basa, Ring aur Tree topology shamil hain. ek 'Grid' network mapping specially aparanparaagat hai aur ise ek standard sthaneey ya vistrita-field topology classification nahi maana jata hai. Internet ki complex, huge architecture ko anivaary roop se ek Tree topology (central basa baikabon ke through level network ko jodne wala ek haaibrida) ke roop me taiyar kiya gaya hai. Ye leaaut nodsa ko hierarchical, manageable vargon me vibhaajit karke paimaane ka effective dhanga se prabandhan karta hai. [Q26–Q29]

- Internet specially paiket Switchinga naam ka technology use karke data ko sthanaantarit karta hai. Badi fileon ko chhote blokon ("paiketa") me vibhaajit kiya jata hai, jo apne gantavy par phira se ikattha hone se pehle vyaktigat roop se ishtatam network pathon ko neviget karte hain. Domain Name System (DNS) anivaary roop se Internet ki sanpark pustika (contact book) ki tarah kaam karta hai. Ye pathaneey saaita naamon (jaise google.com) ko sankhyatmak IP paton me anuvaadit karta hai jinka use computer system kanekt karne ke liye karte hain. Puraane IPv4 paton (jisme 32-Bit lanbaaee ka use kiya gaya tha) ki lessee ko hala karne ke liye, injeeniyaron ne IPv6 ka masauda taiyar kiya. IPv6 standard 128-Bit structure par depend karta hai, jo web se jude devices ke liye adviteey sankhyatmak pahachaanakartaaon ke astronomical roop se bade poola ki allow karta hai. Bluetooth ek sthaneey Protocols ki tarah kaam karta hai jo yooechaeph rediyo frequencyyon use karke kai electronic devices (hedaphona, computer, phone) K Bcha fast, short-range ke vaayarales data brijia banata hai. [Q30–Q33]
- Major specific technical pahaloo opareting frequency hai. Bluetooth strictly 2.4 geegaahartj aaeesaem band ka use karta hai. Iske opposite, standard vaaee-phaaee setup ya to 2.4 geegaahartj spektram ya fast, kam bhida wale 5 geegaahartj band par chala sakate hain. Jaisaa ki pehle light daalaa gaya tha, basic rediyo prasaaran antara ye hai ki bluetooth 2.4 geegaahartj tak hi seemit hai, jabki vaaee-phaaee network me 2.4 geegaahartj ya kam bhida-bhaada wali 5 geegaahartj frequency par kaam karne ka Resilience hota hai. Digital ekeekaran ke liye ek important prayas me, Indian moreaariyon ne teka koraporeshan Google ke saath gathabandhan kiya. Is programme ka purpose kai paaragaman kendron ko taara se jodana tha, jisne 2016 ki shuruaat me muphta Internet ka use start karne ke liye udghaatan test-sthala ke roop me munbaee sentral ko chunaa. Laaee-phaaee (Li-Fi) digital Fileon ko fastee se shuttle karne ke liye timatimaate visible light (aksar standard elaeedee se) ke saath Router dwara use ki jane wali conventional rediyo waves ko pratisthapit karta hai. [Q34–Q37]
- Statement B me government department ke sanbandh me ek tathyatmak truti hai: test information technology ministry ke under ek major electronics braand aur akaadamik injeeniyaron ke saath organized kiya gaya tha, na ki information aur prasaaran ministry. Internet antararaashTreeya level par vividh network ko jodne wali huge architecture ki tarah kaam karta hai. Ye duniya bhar me apaara data sthanaantaran (data transfers) ko sanbhaalata hai, jo water ke neeche Cableon aur high elevation wale Communication Satelliteon ke ek complex sanyojan par depend karta hai. ek veepeeene (VPN) vyapak public Internet par doora ke computeron ko jodne wali ek safe, enkripted surang (tunnel) banata hai. Ye karmachaariyon ko, example ke liye, doora se hi apni Company ke banda Intranet system me safe roop se Login karne ki allow karta hai, opana web par sanbhaavit chhipakar baaten sunane vaalon ko darakinaar karte hue. Portal application ek web-sulabha, intaraiktiv tools hai jo related aur asanbaddh donon application, sevaaen aur linka provide karta hai. [Q38–Q41]
- Talaash (Talaash) Internet par ek Multimedia Portal hai aur saath hi ek optical kairektar rikagnishan (OCR) Software hai jo devanaagaree me C-DAC dwara operated hai. [Q42]

⚠ EXAM TRAP — Q38 — Li-Fi Source Wording

► Source Q38 me multiple false statements mark hain. Li-Fi visible light ko data transmission medium ke roop me use karta hai; exam-year wording ko source context me padhein.

3. Indian Supercomputers, Computer Types, Y2K & Printers

- Software NAYAN ko C-DAC dwara cyber attacks aur data choree ke khataron se bachaav ke liye develop kiya gaya tha. Supercomputers ki ek series 'PARAM' ko Pune me located pragat sanganan development hub (C-DAC) dwara design aur asenbal kiya gaya tha. Indigenously developed pehle Indian Supercomputer ka naam Param-8000 (Param-8000) hai. Param (PARAM) India me C-DAC dwara design kiye gaye Supercomputers ki series ka naam hai. [Q43–Q46]
- Vijay Bhatkar ko Indian Supercomputer ke janak ke roop me jana jata hai. Baarka (BARC) ke computer prabhaag ne samaanaantar prasanskaran takaneekon ke basis par 1991 me Anupam (ANUPAM) project ke under supara computer ka development start kiya tha. ek Personal computer ki word ki lanbaaee usually 32 bits hoti hai jabki Supercomputer ke liye ye 64 bits tak hoti hai. Supercomputer sabhi type ke computeron me sabse fast, sabse large aur sabse mahanga hota hai. [Q47–Q50]
- 1996 ki sthiti ke according, world ka sabse fast computer ASCII reda tha, jiska linapaik (Linpack) demonstration 1 teraaphlopa/sekanda (10¹² opareshan prati sekanda) se zyada tha. Question (2008) ki duration ke dauran, tivehiclehe-1e (Tianhe-1A) ko sabse fast computer maana jata tha. Climate aur jaivik systems me sanbhaavit parivartanon ki Calculation karne ke liye China dwara Supercomputer 'maijika kyooba' banaya gaya hai. 31 maarcha, 2018 tak Pratyush India ka sabse fast Supercomputer tha. Iska unveil janavaree 2018 me Pune located Indian ushnadesheey weather science institute (IITM) me kiya gaya tha. [Q51–Q54]
- India ki raashtrapati shreematee draupadee murmoo ne 13 aktoobara, 2022 ko aaeaeaeete Guwahati me Supercomputer suvidha 'Param kaamaroopa' ka udghaatan kiya tha. KyooBits (qubits) par based ek suparaphaast Quantum computer ek hi question ke bahut saare sanbhaavit uttara ek saath de sakta hai, jo mastishk ki kaaryasystem ki nakala karta hai. Y2K бага, jise mileniyam бага bhi kaha jata hai, computereekrit systems ki Codesinga me ek samasye se related hai jise varsha 2000 ki shuruaat me duniya bhar me tabaahi machaane ka anumaan lagaay gaya tha. Ye question Y2K бага (mileniyama бага) ko refer karta hai jo puraane computer system se related tha, jisme varsha 2000 ko proses karne me samasyaen aa rahi thin. [Q55–Q58]

- Param (PARAM) supara computer ki ek series hai, jise C-DAC, Pune dwara design aur asenbal kiya gaya hai. Ye koi missile nahi hai. W.L.L. Ka matalab Wireless in local loop (Wireless in Local Loop) hai. Aajakal jina computeron ka sabse zyada use kiya jata hai ve Microcomputer hain. Embedded system (embedded system) ek badi mechanical ya electrical system ke andar ek dedicated phankshan wala computer system hai. [Q59–Q62]
- Semiconductor LASER (Semiconductor laser) thosa LASER hain jinka use LASER printer aur seedee/deeveedee pleyar me kiya jata hai. Deskatop printing (DTP) ke liye usually Laser printer ka use kiya jata hai. Daisy wheel printer ek early type ka Impact printer hai jiska invention 1969 me hua tha. [Q63–Q65]

⚠ EXAM TRAP — Q52 — Fastest Supercomputer is Time-Sensitive

- ▶ Source itself Q52 ko outdated mark karta hai. 'World's fastest computer' ranking continuously change hoti hai, isliye exam-year/date ke saath hi memorize karein.

4. WWW, Protocols, Search Engines, Browsers & Network Models

- Ww ka matalab World Wide Web hai. ek English inventoree computer scientist, Tim Berners-Lee ne CERN me kaam karte hue 1989 me World Wide Web invent kiya tha. Tim Berners-Lee World Wide Web (www) ke inventor hain. Ye ek information system hai jo Internet par material share karne me saksham banaatee hai. Is system ke inventor 1989 me Tim Berners-Lee the. [Q66–Q69]
- World Wide Web intaralinkd haaiparatekst dastaavejon aur other resources ki ek information system hai. World Wide Web (WWW) vaishvik level par computeron par rahane wali linka ki gayi information aur resources ka sanplanet hai, jise Internet ke through eksesa kiya ja sakta hai. W3C ka full form World Wide Web Consortium hai. Kisi system ke computereekaran me, vitteey resources, ek ati-modern structure aur janaShakti ke prashikshan ke saath-saath ise karne ki ichchhaaShakti ki requirement hoti hai. Isliye All the four essential hain. [Q70–Q73]
- Cyberspace, Upload aur Modem information technology (inphoteka) aur networking se related shabdaavalee hain. Die gaye options me, Pinaka inphotek shabdaavalee se related nahi hai. Ye India me produced aur DRDO dwara developed ek multiple rocket lonchar hai. Protocols, Login aur Archie sabhi information technology shabdaavalee ka hissaa hain. Protocols (Protocol) niyamon ka ek group hai jo data communication ko control karta hai. Ye communication devices K Bcha ek samajhaute ko represent karta hai. [Q74–Q77]
- Traansamission kontrol Protocols / Internet Protocol (TCP/IP) Internet par network devices ko aapasa me jodne ke liye use kiye jane wale communication Protocols ka ek soota (set) hai. IP pate (IP Addresses) OSI model ki Network Layer (leyara 3) par kaam karte hain. OSI model ke saata-stareeya aarkitekchar me, leyara 1 phijikal leyara hai, leyara 2 data Link Layer hai, aur leyara 3 Network Layer hai. IoT communication ke chaara charan hain Perception Layer, Transport Layer, Network Layer aur Application Layer. Pratyek charan me information sthanaantarit karne ke liye vibhinn vaayarales takaneekon ka use kiya ja sakta hai. [Q78–Q81]
- Network vaataavarana me, 'Network Server' koi specific server category nahi hai, jaisa ki other option hain. Janka e-mail sandesh 'SPAM' (spam) sandesh hote hain jo avaanchhit (unsolicited) hote hain aur service dwara philtar kiye jate hain. Avaanchhit janka e-mail (unsolicited junk e-mails) ka description karne ke liye generally 'SPAM' ka use kiya jata hai. 'SPAM' computer aur Internet se related hai. SPAM Internet par ek hi sandesh ki kai pratiyon ki baadha hai, jo un Peopleon par sandesh thopane ka prayas hai jo anyath ise mila karna nahi chunenge. [Q82–Q85]
- E-mail Bombing neta durupayog ka ek roop hai jisme melaBox ko ovaraphlo karne ya server ko prabhaavit karne ke prayas me kisi pate par badi amount me e-mail bhejana shamil hai. Dakadakago ek Search engine hai jo users ki privacy ko praathamikata dene ke liye kaaphee famous hai. Orkut koi Search engine nahi hai. Ye Google ke svaamitv wali aur operated ek Social Networking website thi jise banda kar diya gaya hai. Baidu INC. ek Chinaee web service Company hai jo Chinaee Search engine sahita kai sevaen provide karti hai. [Q86–Q89]
- Binga (Bing) ek web Search engine hai jiska svaamitv aur sanchaalan Microsoft ke paas hai. Ise 2009 me launch kiya gaya tha. Karentalee (Kurrently) ek nihshulk soshal meediy Search engine hai jo Twitter aur Facebook etc. Jaise soshal meediy pletaphorm se vaastavik time (real-time) ke parinaam provide karta hai. Yahoo, Google aur emaesaen Internet sites (mainly web Portal aur Search engine) hain. Die gaye options me se Yahoo web braaujar nahi hai, ye ek web Portal aur Search engine hai. [Q90–Q93]
- Google Chrome (Chrome), Google dwara developed ek lokapriy web browser hai. Ise pehli baar 2008 me jaaree kiya gaya tha. Web browser ek software application hai jiska use World Wide Web par information tak reach ke liye kiya jata hai. Opera, Vivaldi aur Mozilla Firefox web browser ke example hain. ek web krolara, jise kabhi-kabhi Web Spider ya spaaidarabot kaha jata hai, ek Internet bota hai jo World Wide Web ko systematically braauj karta hai, aama taur par web indeksing ke purpose se. WiMAX ka matalab varlda vaaida intaraoparebilitee phora maaikrovev eksesa hai. Ye ek telecomee technology (communication technology) hai jiska purpose long-range par vaayarales data provide karna hai. [Q94–Q97]
- Analog, Binary code, Chip aur MODE sabhi kisi na kisi roop me computer se jude word hain. Isliye, ye question Contentionita maana ja sakta hai kyonki sabhi computer shabdaavalee ka hissaa hain. [Q98]

⚠ EXAM TRAP — Q98 — Ambiguous Computer Terminology

- ▶ Source Q98 disputed hai; analog/digital, binary code, mode aur chip sab computer-related terms/context me aa sakte hain.

5. Malware, Cybersecurity & Operating Systems

- computer ke field me VIRUS ka meaning 'Vital Information Resource Under Siege' (Vital Information Resources Under Siege) hai. computer virus ek vinaashakaaree Software programme ya Codes ka tukada hai jo users ki information ke binaa computer par loda ho jata hai aur unki ichchha ke viruddh chalata hai. computer virus ek aisaa programme hai jo apni

nakala bana sakta hai aur ek computer se doosare computer me phaila sakta hai, jo usually jaanaboohakar damage pahunchaata hai. computer me phailane wala virus ek vinaashakaaree computer programme (mailaveyara) hai jo human Programmer dwara data ya software ko damage pahunchaane ke liye likhaa jata hai. [Q99–Q102]

- computer virus ek type ka durbhaavanaapoom software programme hai, jise nishpaadit kiye jane par, other computer programon ko sanshodhit karke aur apanaa Codes daalakar khuda ki nakala (replicate) kar sakta hai. computer virus ek Destructive programme hai jise kaaryaAbility ko baadhit karne ya data churaane ke liye engineer kiya gaya hai. computer virus ka meaning Destructive programme se hai jo computer system ko damage pahunchaata hai. Aisaa programme jisme other programon ko sankramit karne, apni pratiyan banane aur other programon me phailane ki Ability hoti hai, virus kehlat hai. Worm (worm) network me freelance roop se phailate hain, jabki virus usually other programme Fileon ko sankramit karte hain. [Q103–Q106]
- Trojan ek type ka malware hai jo khuda ko ek general ya vaidha programme ke roop me chhipaakar users ko durbhaavanaapoom software daaunalod karne ke liye dhokha deta hai. Firewall ek aisee system hai jise kisi private network par anadhikrit access (unauthorized access/attack) ko rokane ke liye design kiya gaya hai. Ye security maanadandon ko pooraa nahi karne wale traiphik ko bloka karta hai. Phaayaravol ek network security device hai jo aane wale aur jane wale network traiphik ki nigaraanee karta hai. Iska main purpose To block unauthorized access to a network hai. Michael Angelo pehle virus me se ek tha jise MS-DOS ko sankramit karne ke liye design kiya gaya tha. Ise punarjaagaran Artkaara Michael Angelo ke janmadin 6 maarch ko active hone aur Hard Drive par data ko nashta karne ke liye programme kiya gaya tha. [Q107–Q110]
- Staksanet (Stuxnet) ek computer Worm (durbhaavanaapoorna programme) hai jo industrial control systems ko lakshit karta hai. Ise specially eeraan ke bushahar nuclear energy sandevice me nuclear senTreephyooja ko damage pahunchaane (saboteja) ke liye design kiya gaya tha. Vaanaakraaee (WannaCry) aur petya (Petya) itihaas ke sabse bade cyber attacks (rainasamaveyara) me se do hain jinhone world bhara ke kai computer network ko prabhaavit kiya. 'itaranalabloo' ek Cyberattacks ka eksaploit tha jiska use vaanaakraaee hamale me kiya gaya tha. Ijaraail ke NSO group ne pegaasas Spyware developed kiya hai, jo world ke sabse aakraamak Spyware me se ek hai. Ye device users ki information ke binaa gupta roop se mobile phone jaise devices me ghusapaith kar sakta hai. MS-DOS (Microsoft Disk Operating System) originally Tim paitarasan dwara likhaa gaya tha aur agasta, 1981 me Microsoft dwara pehli baar jaaree kiya gaya tha. [Q111–Q114]
- MS-DOS gaira-graaphikala command-line (command-aadhaarita) intaraphes (CLI) wala An operating system hai. Isme users computer ko Instructions dene ke liye text command taaipa karte hain. Microsoft dwara 'vindoja 3.0' 1990 me jaaree kiya gaya tha. Isane Graphical User Interface ko kaaphee behatar banaye tha. Vindoj Operating System (Windows operating system) Software development Company Microsoft dwara develop kiya gaya hai. Vindoj Software Microsoft dwara develop kiya gaya hai. options me Microsoft nahi hai, isliye 'None of the above' sahi hai. [Q115–Q118]
- 'UNIX' ek computer Operating System hai jo maltee-yoojara (kai users ko ek saath kaam karne ki suvidhaa) vaataavarana provide karta hai. Ise 1969 me AT&T ke bela laibsa me develop kiya gaya tha. UNIX An operating system hai jiska use server aur kaaryasthal donon me widely kiya jata hai. UNIX Operating System 3 main bhaagon se bana hai: Kernel, shela, aur users command va application. GNU muphta Software ka ek vyapak sanplanet hai, jiska use Operating System ke roop me ya other Operating System ke saath bhaagon me kiya ja sakta hai. [Q119–Q122]

6. Binary Data, Memory Units, LAN, CPU & Hardware/Software

- computer Memory Semiconductor (praaimaree Memory jaise RAM/ROM), maignetik (jaise Hard disk, Floppy Disk), ya optical (jaise seedee aur deevvedee) ho sakatee hai. computer Memory ka main work data aur information ko sanplaneteeet karna hai. Bit ya Binary Digit computer Memory ki sabse chhoti ikaaee hai. Baainaree ek nanbaRing yojana ka description karta hai jisme pratyek anka ke liye kevala do sanbhaavit maana hote hain: 0 and 1. Dashamalav number ko baainaree me badalane ke liye, aapa plesa vailloo ka use kar sakte hain: 8, 4, 2, 1. [Q123–Q126]
- Binary Language me do anka hote hain: 0 and 1. Ye computer dwara samajhee jane wali machine language hai. Yadi kisi Codes ka development isliye kiya jata hai ki jab hama kisi anka ke 4-Bit Codes represent me 1 ko 0 aur 0 ko 1 se badalate hain, to hamen usa anka ka 9 ka poorak (complement) milta hai, ise sva-pooraka Codes (self-complementing code) kaha jata hai. Baainaree ankaganit me, $1 + 0 = 1$, $0 + 1 = 1$, $0 + 0 = 0$ hota hai. ek Byte digital data ki ek ikaaee hai jisme usually aatha Bits (baainaree anka) hote hain. [Q127–Q130]
- 8 bits ke group ko Byte kaha jata hai. Dhvehicle den ki 4 bits ke group ko Nibble kaha jata hai. ek Byte 8 bits ke baraabar hota hai. ek Byte me 8 bits hote hain. [Q131–Q134]
- Ek Byte me 8 bits hote hain, aur 1 Kilobyte (KB) me 1024 bytes hote hain. Bit (Binary Digit ka short roop) computer me data ki sabse chhoti ikaaee hai. ek Bit ka single baainaree maana hota hai, ya to 0 ya 1. computer me kisi word (word) ki lanbaaee ko usually Bits me maapaa jata hai (jaise, 32-Bit ya 64-Bit aarkitekchara). computing me, 1024 Kilobyte (KB) theeka 1 megabyte (MB) ke baraabar hota hai. [Q135–Q138]
- Choonki computer system baainaree svaroopan (aadhaara 2) ka use karte hain, 1 Kilobyte 2 10 Bytesa me anuvaadit hota hai, jo ki 1024 bytes hai. 1 Kilobyte bilkul 1024 bytes ke baraabar hota hai. Listed ikaaayon me, KB (Kilobyte) sabse small hai. PraGati isliye hai: Kilobyte (KB), Megabyte (MB), Gigabyte (GB), aur phira Terabyte (TB). [Q139–Q142]
- Memory ikaaayon ka sabse chhote se sabse bade tak sahi krama hai: Bit (F) -> Byte (B) -> Kilobyte (A) -> Megabyte (C) -> Gigabyte (E) -> Terabyte (D). 1 gigabyte 1024 Megabyte (MB) ke baraabar hota hai, na ki 1024 Kilobyte. (1 megabyte = 1024 Kilobyte). Isliye, option (C) me sanbandh incorrec hai. 1 megabyte (MB) Memory 1024 Kilobyte (KB) ke baraabar hoti hai. 1 megabyte = 1024 Kilobyte. [Q143–Q146]

- Ek Megabyte me 1,048,576 Bytesa hote hain. Isliye, ek miliyan Bytesa lagbhag ek Megabyte ke baraabar hai. Word MB ka universally Mega Bytes ke short roop me use kiya jata hai. Standard yooniCodes mapping sanrachanaatmak roop se pratyek varna ko 16 bits (ya 2 Bytesa) use karke darshaatee hai, jisse ye practical roop se sabhi vaishvik bhaashaaon ke varnon ko samaayojit karne me saksham ho jata hai. Quantum computing me, information ke basic hisse ko 'kyooBit' (kvaantama Bit ka short roop) kaha jata hai, jo klaasikal computing me use kiye jane wale standard 'Bit' ke opposite hai. [Q147–Q150]
- 'kyooBit' word specially Quantum computing ke field se related hai. Ye ek do-raajya (two-state) kvaantama-maikenikala system ki tarah kaam karta hai, jo kvaantam suparapojishan ki wajah se fastee se Calculation karne ki allow karta hai..Net phremavark vaastav me Microsoft dwara mainly vindow ke liye develop kiya gaya tha. ek Local Area Network specially bhaugolik roop se seemit field jaise ek kaaryalay bhavana, skoola ya ghara ke andar computeron ko connect karta hai, jo ek special private network ke roop me poori tarah se kaam karta hai. LAN Local Area Network ka standard short naam hai. [Q151–Q154]
- Eetharanet Local Area Network ke through data ko physical roop se transmit karne ke liye widely apanaay gaya Protocols standard provide karta hai. database shabdaavalee me, ek 'pheelda' anivaary roop se information ki ek specific category (Category of information) ko refer karta hai (jaise 'naam' ya 'aayu' jaisee specific data Specialtaaon wali taalika me ek kolama). Arithmetic Logic Unit CPU (CPU) ke andara essential module hai jo sabhi mathematics sameekaranon aur Logical tulana nirnayon ko sansaadhit karta hai. computer science me, ALU ka meaning 'Arithmetic Logic Unit' (Arithmetic Logic Unit) hai. [Q155–Q158]
- CD 'Compact Disk' (Compact Disc) ka standard short naam hai, jo ek Optical storage maadhyam hai. Magnetic tape aur Magnetic disc (jaise Hard disk drive) donon hi hardware tatva hain jo deerghakaalik use ke liye digital data ki enough amount ko banaye rakhne ke liye famous hain. Isliye donon sahi hain. Bloo-re Disc (BD) ek digital optical Disc data storej phormet hai jise deevdeede (DVD) phormet ko badalane ke liye design kiya gaya tha. Beedee haaee dephinshan (HD) video stora karta hai, jabki deevdeede standard (SD) video ka sammeanign karta hai. (statement 1 sahi hai). Compiler computer hardware nahi hai. Ye ek special Software programme hai jo kisi special programming language me likhe gaye statements ko proses karta hai aur unhe machine language ya "Codes" me convert karta hai jiska use computer ka Processor karta hai. [Q159–Q162]
- Die gaye options me Assembler ek hardware nahi hai. Ye ek Software programme hai jo basic computer Instructionson ko leta hai aur unhe Bits ke paitarn me convert karta hai jiska use computer ka Processor apne basic functions ko karne ke liye kar sakta hai. computer hardware computer ke un physical ghatakon ko refer karta hai jinhen hama dekha aur chhoo sakate hain (jaise monitor, CPU, keyboard, mouse, printer aur Scanner). Excel Microsoft dwara developed ek Application Software hai. An operating system (jaise vindow ya Linux) hardware nahi hai; ye ek system Software programme hai. printer, mouse aur keyboard physical hardware divaais hain. Compiler computer System Software ka ek hissaa hai. Ye programming language ko machine level ki language me convert karta hai jise computer hardware dwara samajha jata hai. [Q163–Q166]
- Internet explorer ek maalikaanaa, kloja-sorsa (closed-source) Software hai jise Microsoft dwara develop kiya gaya hai. Linux, Open office aur Chromium besa (jisakaa Google Chrome use karta hai) ki opana-sorsa Availability hai. Cloud computing me, teen main sarvatr recognition mila freelance service paraten (service layers) hain: Software eja e sarvis (SaaS), pletaphorm eja e sarvis (PaaS), aur inphraastrakchar eja e sarvis (IaaS). "hardware eja e sarvisa" originally laaS ke under aataa hai aur ise apne aapa me ek freelance main service layer ke roop me classified nahi kiya gaya hai. Floppy Disk me ek small saa indeks hola hota hai jiska use phormeting aur reeding functions ke dauran sanplaneteet data sektaron ke shuruaatee point ko refer karne ke liye kiya jata hai. computing aarkitekchar me, ek intarapt rikvest (IRQ) Processor ko bheja gaya ek hardware signal hai jo chala rahe programme ko asthayee roop se rokata hai aur iske bajaay ek special programme ko chalaane ki allow karta hai. [Q167–Q170]
- Operating System (OS) wo Software hai jo computer hardware, Software resources ka prabandhan karta hai aur computer programon ke liye general sevaan provide karta hai. Ye anivaary roop se computer ke basic functions ko control karta hai, jaise ki Memory prabandhan aur work shedyoolinga. Central Processing Unit ko computer ke mastishk ke roop me jana jata hai jahaan moreaansh important Calculationen aur tarka sanchaalan (logic operations) hote hain. Central Processing Unit computer ka sabse important hissaa hai. Yahaan computer mila information ya data ka analysis karta hai. CPU ka full form Central Processing Unit (Central Processing Unit) hai. [Q171–Q174]
- CPU ka full form Central Processing Unit hai. CPU ka meaning Central Processing Unit hai, Control Processing Unit nahi. Other options (e-mail, WWW, ALU) ke statement sahi hain. Kantrol unit (CU) programme Instructionson (instructions) ko diCodes karta hai aur computer system ke sabhi functions ko radioactive aur Instructionsita karta hai. [Q175–Q177]

7. Applications, Computer Generations, IC Chips, Modems & Memory

- Word Processor ek Application software hai jiska use mainly text-aadhaarita Documents banane, sanpaadit karne aur praaroopit karne ke liye kiya jata hai. Microsoft Word ek Word Processor hai, jo ise users ko Documents taaipa karne, phormet karne aur sahejane jaise specific work karne me help karne ke liye design kiya gaya ek application software banata hai. MS Word (emaesa Word) Microsoft dwara banaya gaya ek widely use kiya jane wala Word Processing application hai. Ise paathy information (textual info) ko banane, sansaadhit karne aur prakaashit karne ke liye design kiya gaya hai. Word Processing, spreadsheet Processing aur photo editing ve sabhi programme hain jo users ko specific intaraiktiv work karne me help karne ke liye design kiye gaye hain. Isliye, ve Application Software ke example hain. [Q178–Q181]
- Specific upayogakartaa-end (user-end) functions ko poora karne ke liye Word Processor (jaise MS Word) aur spreadsheet (jaise MS Excel) ka use kiya jata hai, jisse ve Application Software bana jate hain. Excel jaise spreadsheet programme me, data pravishti ke liye moola ikaaee ko 'Cell' (cell) kaha jata hai. Ye ek pankti (row) aur ek kolama ke chauraahe se banata hai. Edoba photoshop (Adobe Photoshop) mainly ek rekhaapunj Graphics sanpaadak hai. Ise ek

major Graphics Software ke saath-saath Multimedia software bhi maana jata hai kyonki iska use chhavi sanpaadan aur Multimedia applicationon ke liye Graphics banane ke liye bade paimaane par kiya jata hai. Isliye option Both (A) and (B) sahi maane gaye. computing me, file (File) information sanplaneteet karne ke liye ek resource hai. ek Word Documents (.doc/.docx), ek video (.mp4), aur ek MP3 audio (.mp3) sabhi fileon ke specific praaroop (format) hain. [Q182–Q185]

- MP3 digital audio ke liye ek famous audio Codesinga praaroop hai. Ye video data nahi le jata hai. MPEG ka full form Moving Picture Experts Group (Moving Picture Experts Group) hai. Ve ISO aur IEC ke under ek work group hain jo audio aur video sanpeedan aur prasaaran ke liye standard established karne ke prabhaaree hain. Pehli generation ke computeron me Vacuum tubes ka istemaal kiya gaya tha. Doosri generation ke computer Vacuum tubeson se hatakar thosa-state Transistors (transistors) ka use karne lage, jisse ve chhote, fast aur zyada worldsaneey ho gaye. Maulik electronic component jisne teesri generation ke computeron ko define kiya, wo Integrated circuit (IC) tha. Is technology ne kai chhote Transistors ko ek Silicon Chip par rakha. [Q186–Q189]
- computeron ko unke vartamaan chhote shape me siCodesne ki Ability Integrated circuit (IC) chips ke invention aur use se sanbhav hui, jo ek chhote se field me laakhon ya arabon Transistors paika karte hain. Integrated circuit (IC) chips Semiconductor material, sabse zyada baara Silicon use karke built hote hain, kyonki unke electrical properties me accurate roop se herapher kiya ja sakta hai. Silicon IC chips ke development me use kiya jane wala major ardhachaalak matter hai kyonki ye highly sthira, prachur amount me hota hai aur ise aasaanee se dosha-free vephars me banaye ja sakta hai. ek computer Chip originally ardhachaalak material ka ek small tukada hai, jo mainly refined A thin layer of silicon use karke taiyar kiya jata hai. [Q190–Q193]
- IC Chip banane me shamil sabsatret aur baad ki starit doping processon me maulik roop se Silicon (aur iske oksaaida) ka use kiya jata hai. IC chips Silicon ke bane hote hain, jo sarkiTree ke liye ardhachaalak basis ke roop me kaam karte hain. Jaisaa ki kai questionon me establish kiya gaya hai, Silicon IC chips ke liye standard aadhaarabhoot material hai. Integrated circuit ke andara Semiconductor Chip Silicon ki banee hoti hai. [Q194–Q197]
- Silicon ne kraanti laa dee hai aur modern computing age ke dauran computer chips ke production me lagaataar define matter bana hua hai. (nota: Gallium Arsenide ka use kuch special haaee-speeda chips me hota hai, lekin widely Silicon hi main basis hai aur is question ke context me Silicon correct answer maana gaya hai). Integrated circuits (ICs) ko unke ekeekaran paimaane dwara classified kiya jata hai - jiska anivaary roop se meaning hai Chip par paika kiye gaye Transistors ka ghanatv ya number. ek Modem (modulator-demodulator) ek hardware divaais hai jo digital praaroop se data ko Analog praaroop me (Telephone laainon par traansamission ke liye) aur Analog se digital me (data mila karne par) convert karta hai. Modem (Modulator/Demodulator ka short roop), ek hardware divaais hai jo digital signals into analog signals aur analog signals into digital signals parivartit karke computer ko Telephone laainon par information bhejne aur mila karne ki allow karta hai. [Q198–Q201]
- Ek Modem modyooleshan (digital data ko Analog signal me parivartit karna) aur deemodyooleshan (aane wale Analog signal ko vaapas digital data me parivartit karna) donon ke liye jimmedaar hai. ek Modem digital computer signal aur conventional Telephone laainon dwara use kiye jane wale Analog signal K Bcha pula ki tarah kaam karta hai, jisse vaishvik data communication ki anumati milti hai. Modem wo device hai jise specially Internet aur network eksesa ke liye computer (jaise Microcomputer) ko Telephone laainon se jodne ke liye design kiya gaya hai. Choonki data do dishaaon me bahata hai, isliye ek Modem ko computer se digital output ko phone line (modyooleshana) ke liye Analog signal me badalana hota hai, aur aane wale Analog signal ko computer ke liye vaapas digital data (deemodyooleshana) me badalana hota hai. [Q202–Q205]
- Modem wo accurate device hai jo standard Telephone laainon par computer data ko traansamitebal aavegon (impulses) me anuvaad karne ke liye jimmedaar hai. Modem ka primary work computer (jo Digital signal ka use karta hai) aur ek Telephone line (jo Analog signal le jata hai) K Bcha ek intaraphes ke roop me work karna hai. Phone laainon ke through FAX aur e-mail bhejne me prayukt information signal ke roop me Analog aur digital donon signal shamil hote hain, kyonki computer data ko digital roop se proses karte hain lekin Telephone laainen ise Modem ke through Analog vevaphorm me transmit karti hain. Decoding wo essential process hai jisadwar riseevor source dwara bheje gaye enCodeseda prateekon ya signalon ka anuvaad aur vyakhy karke vaapas samajhne yogya information me badalt hai. [Q206–Q209]
- Router, Switch aur haba sabhi networking divaais hain jinka use computer ya network ko ek saath jodne ke liye kiya jata hai. ek Router sabse aama networking divaais hai jo specially vibhinn computer network (Internet par traiphik nideshak ke roop me work karna) K Bcha data paiket ko agreshit karta hai. ek Repeater ek networking component hai jo network ki lanbaaee badhaata hai. Ye ek signal mila karke aur ise aage badhane se pehle ise badhaa/punarjeevita (amplifying/regenerating) karke receive karta hai, jisse long-range par signal ke lessjor hone ko rokaa ja sakta hai. Cellerona, pentiyam aur core sabhi intela dwara built computer Processor (CPU) ke highly lokapriy family (seereeeja) hain. [Q210–Q213]
- ROM ka meaning reeda-onalee Memory (Read-Only Memory) hai. ek baara jab data ko ROM Chip par likha diya jata hai, to ise generally badala ya hataay nahi ja sakta hai aur kevala padhaa ja sakta hai. ROM (reeda-onalee Memory) ko computer ki sthayee internal Memory maana jata hai. computer banda hone ke baad bhi ROM me data ya information nashta ya mitaaee nahi jati hai. Gaira-vaashpasheela computer Memory bijalee hatane par bhi data barakaraar rakhti hai. ROM (reeda-onalee Memory) computer borda me built iska primary example hai. computing me, Memory Address ek specific sankhyatmak pahachaanakarta hai jiska use CPU dwara computer ki Memory ke andar ek accurate sthana par sanplaneteet data ka detect lagaane aur usa tak reach ke liye kiya jata hai. [Q214–Q217]
- CD-ROM ka meaning Compact Disk reeda-onalee Memory (Compact Disc Read-Only Memory) hai. Ye ek optical Disc praaroop hai jisme data sthayee roop se mudrit hota hai aur ise mitaay nahi ja sakta hai. RAM raindama-eksesa Memory (Random Access Memory) ka short roop hai. Ye ek asthayee kaaryakshetr hai jahaan computer un data aur programme ko loda karta hai jinka wo active roop se use kar rahaa hai. RAM (Random access memory) asthir (volatile) hai, jiska meaning

hai ki ise data rakhne ke liye nirantar electrical Shakti ki requirement hoti hai. Jab computer banda kar diya jata hai, to RAM ka saara data mita jata hai. RAM Hard disk store ki tarah nahi hai. Hard disk sthayee (permanent), gaira-vaashpasheela storage provide karti hai, jabki RAM specially CPU ke active use ke liye asthayee, vaashpasheel aur behada fast eksesa store provide karti hai. Isliye statement B true nahi hai. [Q218–Q221]

- Random access memory ek computer system ki main ya primary Memory (primary memory) hai, jo CPU ke saath active roop se intaraikt karti hai. Solid State Drive, Hard disk aur pen drive sabhi sekendaree (external aur sthayee) Memory ke roop hain. RAM primary (primary) Memory ki tarah kaam karta hai, jo CPU ko kaam karne wale data tak immediate reach me enable karta hai. Kaisha (Cache) Memory ek atyant fast, small Memory component hai jo CPU ke kareeb ya uske andara integrated hota hai. Ye usa data aur Instructionson ko baphar karta hai jinhen CPU sabse zyada baara eksesa karta hai, jo haaee-speeda CPU aur apekshaakrit dheemee main RAM K Bcha Gati ke antara ko paatata hai. Specific system Memory K Bcha, kaisha (Cache) me sabse kam access time (shortest access time) aur hightam Bandwidth hoti hai, jo niyamt RAM ya magnetic storage ke comparison me bahut fast pratikriy karti hai. [Q222–Q225]
- Kaisha (Cache) Memory specially baara-baara use kiye jane wale data ko CPU ke liye immediate upalabdh rakhakar computer Processing me fastee laane ke liye dedicated hai. Kaisha (Cache) Memory yahaan listed sabhi options me sabse fast Memory Unit ki tarah kaam karti hai, jise directly CPU aarkitekchar par ya uske bagala me banaya gaya hai. Flash Memory (SSD, pen drive me prayukta) solid-state aur energy-kushala hai, lekin paaranparika, revolving mechanical Hard disk draaiv (HDD) ke comparison me iski prati Gigabyte storage laagat aama taur par zyada hoti hai. Isliye, statement (C) tathyatmak roop se incorrect hai. [Q226–Q228]

⚠ EXAM TRAP — Q184, Q210 & Q211 — Multiple/Disputed

- ▶ Adobe Photoshop ko source graphics aur multimedia dono categories me accept karta hai.
- ▶ Router, switch aur hub sab connecting devices hain; Q210 flawed/bonus-type source item hai.
- ▶ Networks ke beech packets forward karne ka standard answer router hai, lekin Q211 options/context ki wajah se source disputed mark hai.

8. Programming, Domains, Communication, AI & Digital Technologies

- COBOL ek high-stareeya computer programing language hai. Iska naam komana bijanes oriented laingvej (Common Business Oriented Language) hai, jise mainly vyavasaaya, vitta aur prashaasanik systems ke liye develop kiya gaya hai. ORACLE (ORACLE) ek highly famous maltee-model rilesanal database mainejament system hai, jise aama taur par database software ke roop me classified kiya jata hai. Basic (BASIC), see (C) aur phoratraan (FORTRAN) universally recognition mila programming bhaashaaen hain jinka use computer Software ko Codesinga karne ke liye kiya jata hai. 'FAST' ko computer Language ke roop me recognition mila nahi hai. Java, C++ aur PASCAL programming bhaashaaen hain. ROM (reedaa-onalee Memory) computer ke andara ek physical hardware component (Memory Chip) hai, ye koi programming language nahi hai. [Q229–Q232]
- Paint brush (MS Paint) ek sarala Graphics penting Application Software hai, jabki Basic, C++ aur Java high-stareeya programming bhaashaaen hain. Listed All the four option vaastavik computer programing bhaashaaen hain. Kobara (Cobra) ek objekta-orienteda language hai, paayathan aur Java aworldsaneey roop se lokapriy general-purpose bhaashaaen hain, aur skvirai (Squirrel) ek halkaa skripting language hai. COBOL, PASCAL aur BASIC sabhi high-stareeya programming bhaashaaon ke klaasik example hain, jinhen low-stareeya (low-level) machine Codes ke comparison me manushyon dwara zyada aasaanee se padhane aur likhane ke liye design kiya gaya hai. Phoratraan (FORTRAN - phormoola traansaleshana) ko clearly 1950 ke decade me IBM dwara teevra sankhyatmak Calculation aur scientist computing ke liye develop kiya gaya tha. [Q233–Q236]
- Basic (BASIC) originally ek procedural programming language hai. Ye functions ko nishpaadit karne ke liye statements (statements) ke raikhik sequence (linear sequence) par depend karta hai. Paanchaveen generation ki bhaashaaen (5GL) charana-dara-charana Algorithms par highly depend hone ke bajaay tarka aur baadhaa-aadhaarita programming ka use karti hain. Prolog ek major example hai, jo kritrim buddhimatta (AI) research aur Specialjnya systems (expert systems) se depression se linked hai. JSP Java Server Pages ka short roop hai. Ye ek server-saaida programming technology hai jo Software devalapars ko HTML aur XML ke basis par Gatisheela roop se utpann web peja banane me help karti hai. Assembler ek specific upayogita (utility) program hai jo asenbalee language (jo pathaneey nimoniks ka use karta hai) me likhe gaye Codes ko shuddh baainaree machine language me anuvaad karne ke liye jimmedaar hai jo computer Processor dwara nishpaadan yogya hai. [Q237–Q240]
- Machine language computer ki mauilik language hai. Isme kevala do anka 0 and 1 shamil hote hain. computer ki Machine language Boolean Algebra par based hai. Ye mathematics ka wo roop hai jo statements aur unke booliyan maanon se related hai. Sabhi digital computer 1 aur 0 ki baainaree system aur 1850 ke decade me English inventoree ganitajny jorja boola dwara determine tarka (logic) ke niyamon par depend karte hain. '.us' domain naam The United States of America ko refer karta hai aur '.edu' ka meaning shiksha hai. Isliye, .edu.us ek American academic institute ka domain naam hai. [Q241–Q244]
- Organisation ki website ka.com bhaaga lessrshiyal (commercial) ka short roop hai. Ye Internet ki domain naam system me ek sheersha-stareeya domain (TLD) hai. Domain naam me last point ke baad wale bhaaga ko topa levala domain kaha jata hai, jaise www.cnn.com. Isme ".com" topa levala domain hai. Proksee server teeseeppee/aaeepee detect (TCP/IP address) provide nahi karta hai (ye usually deeechaseeppee (DHCP) server ka kaam hai), isliye statement 1 incorrect hai. LTE ko aama taur par 4G (3G nahi) ke roop me vipanan kiya jata hai, aur VoLTE advanced 4G voice traansamission se related hai. Isliye, statement 1 incorrect hai. [Q245–Q248]
- Visible light communication (VLC) ek chhoti distance ka (short-range) optical vaayarales communication hai, jo statement 2 ke opposite hai. Isliye, statement 2 incorrect hai. Niyara pheeld Communication (NFC) devices ko bahut short-range par,

usually 4 senteeMetre ya usase short-range par information share karne me enable karta hai, na ki ek Metre tak. Isliye, statement 2 incorrect hai. Miniaturized-category ke device (Short Range Devices - SRD) kam sancharit Shakti ke saath kaam karte hain aur isliye unki seemaa kam hoti hai. Statement 1 and 2 ne anivaary roop se AR aur VR ki paribhaashaaon ki adalaa-badalee kar dee hai. Aabhaasee vaastavikata (VR) ek simyuleted vaataavaran banaatee hai jo physical world ko poori tarah se banda kar deti hai. Sanvardhit vaastavikata (AR) vaastavik world ke parivesh par computer generated chhaviyon ko projekt karti hai. [Q249–Q252]

- John McCarthy ko 'aartiphishiyala intelijens (AI) ka janak' maana jata hai. unhone 1950 ke decade me 'aartiphishiyala intelijensa' word gadhaa tha aur is field me ek major pioneer vyakti the. 1950 me Alan Turing dwara developed The Turing Test, jiska purpose ye moolyankan karna tha ki kya koi machine human ke baraabar ya usase aprabhedy buddhimaan behavior display kar sakti hai, ko originally "imiteshana gema" (Imitation Game) kahaa jata tha. Statement A sahi hai: Machine Learning (Machine learning) aartiphishiyal intelijens ka ek upasamuchchay hai. Machine Learning (ML) aartiphishiyal intelijens ki ek special branch hai jo computeron ko data se svaayatt roop se seekhane aur demonstration me improvements karne me saksham banane par kendrit hai. [Q253–Q256]
- AI baayyas (pakshapaata) taba hota hai jab koi AI system pakshapaatee parinaam generate karta hai. Ye mainly pakshapaatee prashikshan data, Algorithms design me technical issues, ya unhe banane wale Peopleon (People) ke poorvaaplaneton se utpann hota hai. Jab ye question taiyar kiya gaya tha (2020), taba AI energy Grid ko anukoolit karne (1), chikitsa chhavi vishleshana/roga nidaan (3), aur text-too-speecha technology (4) me highly effective tha. FPGA ka full form 'Field Programmable Gate Array' (Field Programmable Gate Array) hai. Ye ek type ka Semiconductor IC hai jise development ke baad graahak ya designar dwara phira se programme ya apadet kiya ja sakta hai, jo AI functions me lacheele hardware tvaran (acceleration) ki allow karta hai. Air indiya ne graahak service ko badhane aur vividh questionon ka kushalataapoorvak utara dene ke liye Microsoft ke ejyora OpenAI (Azure OpenAI) service dwara operated 'Maharaja' (Maharaja) naam ka ek jenareativ eaaee varchual ejenta pesha kiya. [Q257–Q260]
- Chatjeepeetee (ChatGPT) ek bade language model (LLM) par based chatbot hai jise OpenAI dwara develop kiya gaya hai aur 2022 ke end me jaaree kiya gaya hai. Ye users ko AI ke saath intaraaktiv vaartaalaap karne me enable karta hai. Election Commission of India (ECI) ne raajaneetik dalon aur ummeedavaaron ko paaradarshee aur kushalata se chunaav prachaar Gatividhiyon ke liye anumati mila karne ki suvidha provide karne ke liye advanced 'suvidhaa 2.0' mobile aipa launch kiya. U.S.B. Ka full form 'Universal Serial Bus' (Universal Serial Bus) hai. Ye ek udyoga standard hai jo computer aur paridheey (peripheral) devices K Bcha connection, communication aur bijalee aapoorti ke liye Cable, kanektar aur Protocols ke liye vilInstructions established karta hai. ek Port computer aur external ya internal devices (eksesareeja) jaise ki mouse, printer, ya keyboard K Bcha ek specific connection point ya intaraphes ki tarah kaam karta hai. [Q261–Q264]
- Password varnon (characters) ki ek stRing hai jiska use users ki pahachaan satyapit karne ke liye kiya jata hai, jisse computer system ya network par anadhikrit access (unauthorized access) ko rokaa ja sakta hai. computing me, 'Finger' (Finger) ek networking Protocols hai jiska use kisi specific network par users ke baare me information ka detect lagaane ke liye kiya jata hai, jaise ki unakaa asalee naam, e-mail detect ya logina sthiti ki jaanch karna. Pablik ki inphraastrakchar (PKI) Digital security infrastructure (digital security infrastructure) ka ek abhinn anga hai. Isme information ko safe roop se sthanaantarit karne ke liye digital pramaanapatron (digital certificates) aur pablika-ki (public-key) encryption ka nirmaana, prabandhan aur use shamil hai. Digital Signature ek ganiteey technology hai jiska use kisi sandesha, software ya digital Documents ki praamaanikata aur Integrity ko maanya karne ke liye kiya jata hai. Ye mainly ye sunishchit karne wale electronic hastaakshar ki tarah kaam karta hai ki material aparivartit rahe (statement 3 sahi hai). [Q265–Q268]
- The Information Technology Act, 2000 ne India me e-Commerce aur electronic Documentseekarana ke liye kaanoonee dhaancha establish kiya. Isane Electronic Record, Digital Signature ko kaanoonee recognition provide ki aur Certifying Authority ki bhoomika ko define kiya. Isliye sabhi option sahi hain. Statement 1 sahi hai: BHIM (bhima) aipa UPI aaeedee, phone nanbar ya UPI-sakshama Banks khaate se jude kyooaar (QR) Codes ko skaina karke tvarit dhana hastaantaran ki allow karta hai. Digital indiya ka main phokas graameen kshetron me network kanektivitee ka expansion karke, public vaaee-phaaee provide karke aur government sevaon ko electronic roop se vitarit karke India ko digital roop se sashakt samaaj me badalana hai. Isliye, statement 3 sahi hai. India government ne digital public vastuen (public goods) established karne ke liye opana-sorsa praudyogikiyon par bahut zyada jora diya hai. Listed All the four pletaphorma—aarogya setu, kovina, dijeelokar aur deekshaa—antarnihita opana-sorsa digital aarkitekchar par depend karte hain, jo paaradarshita aur skeLabel ekeekaran ko protsaahit karte hain. [Q269–Q272]
- Statement 1 incorrect hai: basis adhiniyam ke according, basis number strictly pahachaan ka pramaan hai, naagarikata ya adhivaas (domicile) ka pramaan patra nahi. Khule (Open) API ka matalab hai ki Software devalapars aise Codes likha sakate hain jo pahachaan satyapan ke liye basis database ke saath directly intaraphes karne ke liye hardware (jaise phone ya pointa-opha-Cell machine) ki ek detailed vividhata ki allow karta hai. Isliye statement 1 sahi hai. Statement 1 sahi hai: blokachen mauulik roop se ek vikendreekrit (decentralized), vitarit public bahikhaate (ledger) ki tarah kaam karta hai jo nireekshan ke liye sulabh hai lekin ekatarapha herapher ke pratirodhee hai. Web 3.0 ek vikendreekrit Internet ki parikalpana karta hai. Blokachen jaisee vitarit khaata praudyogikiyon par depend hokara, ye Shakti ko kendreekrit technical diggajon se doora kar deta hai. [Q273–Q276]
- Bele II (Belle II) prayog ek major *particle physics* prayog hai jo B mesana ki jaanch kar rahaa hai, na ki aartiphishiyal intelijensa. (jodee 1 incorrect hai). CRISPR-Cas9 ek kraantikaaree *jenetika engineering/jeena-editinga* tools hai, jo paartikal phijiks se related nahi hai. (jodee 3 incorrect hai). [Q277–Q278]

9. Devices, E-Governance, Multimedia, Video, E-mail & Mobile ICT

- computer computing, determine niyamon ke according fastee se data ko proses karne aur parinaam output karne me highly kushal hote hain. computer shabdaavalee me, computer programme ya system me koi truti, khaamee ya dosha jo use incorrect ya apratyashit parinaam dene ka kaaran banata hai, use widely "baga" (Bug) kaha jata hai. MICR 'magnetika

inka kairektar rikognishana' ka short roop hai. O.M.R. 'Optical Mark Reader' (ya rikognishana) ka short roop hai. Ye computer system me data enter karne ki ek method hai. [Q279–Q282]

- OMR (Optical Mark Reader) technology ka use mainly bulabule (bubbles) se bhare vastunishth type ke utara patron ko skaina karne aur jaanchane ke liye kiya jata hai. I.R.C. Ka meaning Internet Relay Chat hai. Ye ek application leyara Protocols hai jo Peopleon ko text-aadhaarita vaataavarana me vaastavik time me Ek-dosare ke saath sanvaad karne ki allow karta hai. 2G 'dosri generation' (2nd Generation) ki vaayarales Telephone technology ka short roop hai. Ye hamen text aur chitra sandesh bhejne me enable karta hai aur isme basic Internet Abilityen hoti hain. CAD (computer ededa designa) Software ka use aarkitekta, injeeniyara, draaphtara, Artkaara aur other Peopleon dwara accurate chitra ya technical chitra banane ke liye kiya jata hai. [Q283–Q286]
- GIF ka meaning 'Graphics intarachenj phormeta' (Graphics Interchange Format) hai. Ye ek computer file svaroop hai jiska use chhaviyon aur sarala enimeshan ke liye kiya jata hai. 3D printing word me vibhinn type ki processen shamil hain jinamen material ko computer control ke under jodaa ya jamaay jata hai taaki tri-aayamee (3D) objekt banaye go saken, jo usually layer-dara-layer hote hain. E-Governance, jiska meaning hai 'electronic gavarnensa', dakshata badhane ke purpose se technology operated shaasan provide karne ke liye government ke vibhinn staron par information aur Communication technology (ICTs) ka use kar rahaa hai. Government sevaaen provide karne ke liye shaasan me computer ke use ko E-Governance ke roop me jana jata hai. [Q287–Q290]
- E-Governance ke chaara major stanbh People, process, technology aur resource hain. Ye stanbh saamoohik roop se shaasan ke digital parivartan ka sammeanign karte hain. Gaartana ka E-Governance model ek chaara-charaneeya model hai jo ee-government ke development ko measure karta hai. Charan hain: Sushaasan (good governance) mila karne ke liye government me aaeeseetee (ICT) ke application ko E-Governance kaha jata hai. computer fast Gati se Calculation karta hai. ek Microcomputer prati sekand laakhon Calculationen kar sakta hai. [Q291–Q294]
- Multimedia wo field hai jo text, Graphics, draainga, chalatee chhaviyon (video), eneemeshana, audio aur kisi bhi other meediy ke computer-radioactivea ekeekaran se related hai jahaan hara type ki information ko digital roop se darshaay ja sakta hai. Multimedia applicationon me kai type ke meediy (text, audio, video, Graphics) shamil ho sakate hain. Multimedia system ki primary Specialtaa ek saath material vitarit karne ke liye ek se zyada type ke meediy ka use karna hai. computing me, Multitasking (multitasking) ek nishchit duration me kai functions ya applicationon ko ek saath nishpaadit karne ki concept hai. Telefax ke through, koi Telephone laanon ki help se ek sthana se dosare sthana par Graphics aur paathy information communicationit kar sakta hai aur unhe screen par padha sakta hai. [Q295–Q298]
- Video mela ek former-rikorda kiye gaye video ko e-mail ke roop me bhejne ki ek system hai. Graphics, Video clips aur Video messages un sabhi material ke example hain jinhen video mela ke through bheja ja sakta hai. Video conferencing ka meaning audio aur video data ko ek saath transmit karne ke liye computer network use karke vibhinn saaton par do ya do se zyada pratibhaagiyon K Bcha sammelan organized karna hai. Video conferencing telecom praudyogikiyon ke ek seta use karke video kola ka sanchaalan hai jo do ya do se zyada sthanon ko ek saath do-taraphaa video aur audio prasaaran dwara sanvaad karne ki allow karta hai. Video konphrens ke liye, laaiva video capture karna essential hai, jiske liye Video camera ya Webcam ki requirement hoti hai. [Q299–Q302]
- Video conferencing system ke liye essential component Video camera, Display device aur Microphone hain. Telephone ki requirement nahi hai kyonki audio Internet aur maaikrophona/speekara ke through transmit hota hai. E-mail ka full form 'Electronic Mail' (Electronic mail) hai. ek maanya e-mail detect do bhaagon se bana hota hai, ek users naam (Username) aur @ prateek dwara alaga kiya gaya ek domain naam (jaise, username@domainname.com). Kisi bhi e-mail pate me @ chinha ke baaen direction ka bhaaga users naam (username) hota hai aur @ chinha ke daaen direction ka bhaaga domain naam hota hai. Isliye die gaye pate me, ITdesk.info domain naam hai. [Q303–Q306]
- Hotmail, Rediffmail, Yahoo aur jeemel sabhi famous muphta e-mail service pradaata hain. E-mail ataichament e-mail sandesh ke saath bhejee gayi ek digital file hai. Jina fileon ko sanlagn kiya ja sakta hai unamen Documents, text fileen, Music, audio fileen, video fileen, photo etc. Shamil hain. Aapa ek e-mail me lagbhag kisi bhi type ki file sanlagn kar sakte hain, jisme text, audio aur video fileen shamil hain, jab tak ki file ka shape e-mail pradaata dwara determine seemaa ke andar ho. 'data' (data) word ka ekavachan roop 'Datum' (Datum) hai. [Q307–Q310]
- 'Blog' ek web peja hai jo public roop se sulabh vyaktigat jarnal ki tarah kaam karta hai. 'Blog' word 'web' (web) aur 'loga' (log) do shabdon ka sanyojan hai. CPU Instructions chakra me, Fetching Instructions mila karna hai, Decoding ise samajhana hai, aur Executing (executing) aadeshon ko poora karne ki vaastavik process hai. Software yoojar intaraphes me, menoo me related commands (aadeshon) ki ek list hoti hai jise users kriyaon ko nishpaadit karne ke liye chuna sakate hain. Bookmark apne browser me apni pasandeeda website ko saheja kar (save) usa tak sheeghrata se reach ka ek method hai. Ye baad ke sandarbh ke liye websiteon ko sahejane ka ek upayogee aur fast method hai. [Q311–Q314]
- Varchual Memory An operating system ki Specialtaa hai jo computer ko asthayee roop se Random access memory se Disc storej me data traansaphar karke physical Memory ki lessee ki bharapaaee karne me saksham banaatee hai. ICT ka full form 'Information & Communications Technology' (Information and Communications Technology) hai. Ye un praudyogikiyon ko refer karta hai jo telecom ke through information tak access provide karti hain. SIM ka full form Subscriber Identity Module (Subscriber Identity Module) hai. Ye ek small kaarda hai jisme mobile network graahak ke khaate ki information hoti hai. SMS ka matalab 'Short messaging service' (Short Messaging Service) hai. Iska use mobile phone dwara sandeshon ke aadaana-pradaana ke liye kiya jata hai. [Q315–Q318]
- Usa time ke general teleekom maanadandon ke according, ek pikchar maisej mobile ke private inaBox me 30 Dayson tak raha sakta hai. Ise zyada dinon tak safe rakhne ke liye ise sandesh Box (message box) me sahejana (save) padata hai. Option (B) me diya gaya yugma sahi matched nahi hai. Endroid (Android) voice input system nahi hai; ye Google dwara developed ek mobile Operating System (OS) hai, jise mainly tacha-skreena mobile devices ke liye design kiya gaya hai. [Q319–Q320]

EXAM TRAP — Q319–Q322 — Legacy Mobile Facts

► Picture-message retention, Android 6.0 naming aur iPhone 4S specifications historical/source-period facts hain; current mobile-tech facts ke roop me use na karein.

10. Mobile Platforms, Cybercrime, Data Security & Computer Networks

- Endroid 6.0 'Marshmallow' endroid Operating System ka chhatha major version tha. Mae 2015 me pehli baar iska unveil kiya gaya aur ise official taur par aktoobar 2015 me jaaree kiya gaya tha. iPhone 4S me 512 MB RAM (640 MB nahi) ke saath dual-core Apple A5 Processor tha. Isme 200 ghante ka staidabaay taaima (300 ghante nahi), 3G par 8 ghante ka toka taaima tha aur Siri (Siri) naam ka ek new voice rikagnaaijing asistent pesha kiya gaya tha. Apni shikaayat ko nikaalane ke liye website ka use karna 'Cyber venting' kehlat hai. Denial of Service (DOS) ek type ka hamala hai jahaan haikars vaidha users ko website tak reach se rokane ka prayas karte hain. [Q321–Q324]
- Cyber kaanoon shabdaavalee me, 'DOS' ka meaning 'Denial of Service' (Denial of Service) hai. Ye network service ko baadhit karne ke liye design kiye gaye hamale ko refer karta hai. (nota: general computing me, DOS ka meaning Disk Operating System hota hai). Distribyooted dinayala-opha-sarvisa (DDoS) A cyber attack hai. Ye kai alaga-alaga sourceon (computeron) se ek saath amaany anurodh bhejake lakshit server ko abhibhoot karne aur network ke general traiphik ko baadhit karne ka ek durbhaavanaapoom prayas hai. Hacking (Hacking) kisi system ya resources tak anadhikrit access mila karne ke liye lessjoriyon ka phaayada uthaane ki process hai. Ye na to kevala discovery hai aur na hi security provide karna hai. Isliye 'None of these' correct answer hai. Lokapriy sanskriti me, computer haikar wo vyakti hai jo lessjoriyon (bugs) ka phaayada uthaane ya computer system me sendha lagaane ke liye technical jnyaan ka use karta hai, jo aksar vyaktigat laabha ke liye durbhaavanaapoom iraade se computer security ka ullanghan karta hai. [Q325–Q328]
- Ek Cracker (ya blaika haita haikara) wo vyakti hota hai jo gopaneey data tak reach ya durbhaavanaapoom kshati pahunchaane ke liye computer ya network system me sendha lagaata hai. Blaika haita haikar (Black hat hacker), jise kabhi-kabhi cracker bhi kaha jata hai, wo hota hai jo binaa kisi moreaar ke computer security ko todta hai aur system ko jaanaboohakar damage pahunchaane ya dhokhaadhaadee karne ke liye technology ka use karta hai. ek Cracker ya blaika haita haikar kisi computer system ya network me anadhikrit access receive karta hai, important data ko nashta karta hai, aur vyaktigat ya durbhaavanaapoom kaaranon se apne lakshyon ke liye samasyaen paidaa karta hai. ek varchual keyboard physical kunjiyon (physical keys) ke bajaay screen intaraphes ke through input ki allow karta hai. Ye hardware ya software Keyloggers (jo aksar Trojan ya Spyware se jude hote hain) ko physical keestros ko record karne se rokata hai, isliye Password theft se bachata hai. [Q329–Q332]
- Hacking, Cyber Stalking (onalaaina peechha karna) aur Denial of Service (DOS) hamale sabhi digital network par ki jane wali avaidh Gatividhiyan hain, jo unhe sabhi recognition mila type ke cyber aparaadh banaate hain. Online Chatting ek saamaanya, kaanoonee digital communication Gatividhi hai. Phishing, Cyber Stalking aur Identity Theft All of these durbhaavanaapoom Gatividhiyan hain jinhen cyber aparaadh ke roop me classified kiya gaya hai. Phishing hamala bhraamak (dhokhebaaja) e-mail ya text sandeshon ke roop me aataa hai jo peedit ko sensitive information (jaise Password) share karne ya durbhaavanaapoom fileen daaunalod karne ke liye lubhaane ke liye design kiye gaye hain. Phishing (Phishing) hamale users ko ek worldsaneey ikaaee (jaise Banks ya badi Company) ke roop me prachchhann hokara logina kredenshiyal ya kredit kaarda nanbar jaisee sensitive information pratat karne ke liye dhokha देने के liye soshal engineering ka use karte hain. [Q333–Q336]
- E-mail Spoofing jaalee preshak (sender) pate ke saath e-mail sandeshon ka development hai. Ye ek aisee technology hai jiska use aksar phishing aur SPAM abhiyanon me kiya jata hai kyonki Peopleon ke e-mail kholane ki zyada sanbhaavana hoti hai jab unhe lagata hai ki ye kisi vaidha source dwara bheja gaya hai. Cyber aparaadh ka ek major define pahaloo ganbhir jnyaan ki khaaee hai: aparaadhiyon ke paas system ka phaayada uthaane ke liye special computer kaushal (skills) hote hain, jabki peedit usually ina technical lessjoriyon se anabhijny (ignorant) hote hain. Vyapak cyber security me viniyamak Compliance (regulatory compliance), internal aur external donon khataron se bachaava, khatare ka detect lagaana aur active rokatham ke upaaya shamil hain. The Information Technology Act, 2000 (IT Act) India me cyber aparaadh aur electronic komars se nipatane wala primary kaanoon hai. Ye electronic data intarachenj ke through kiye gaye lenaden ko kaanoonee recognition provide karta hai. [Q337–Q340]
- Hub government dwara jaaree information technology niyama, 2013 ke under, service pradaataaon, bichauliyon, data kendron aur nigamit nikaayon (body corporates) ke liye CERT-In ko cyber security ghatanaaon ki riPort karna kaanoonee roop se anivaary hai. Cyber beemaa digital ghatanaaon (System ki bahaalee, salaahakaara, kaanoonee bachaava) se hone wale amoort damage ko kavara karta hai. Ye usually physical kshati ya hardware ke pratisthapan ko kavara nahi karta hai (jaise ki kisi badamaash dwara physical roop se tode jane par new computer khareedanaa). Isliye, statement 2 incorrect hai. Janaral data protekshan reguleshan (GDPR) The European Union (EU) aur European aarthik field (EEA) me data security aur privacy par The European Union ke kaanoon me ek niyama hai. Privacy wo principle hai jo sensitive information ko private rakhne aur anadhikrit access (unauthorized access) se safe rakhne ke importance ko rekhaankit karta hai. [Q341–Q344]
- computer me Basic inaputa-outputa system (BIOS) ek firmware hai jiska use booting process ke dauran hardware inishiyalaaijeshan karne ke liye kiya jata hai. Ye motherboard par gaira-vaashpasheela (non-volatile) ROM chips par sanplaneteet hota hai. computeron ko ek saath jodakar networking unki upayogita aur information access Abilityon ko kaaphee badhaa deti hai. Haalaanki, devices ko jodne se bhedyata aur khataron (jaise virus aur Hacking) ki sanbhaavana bhi badha jati hai. Resilience general sanchaalan me doshon aur chunautiyon ke baavajood service ka ek sveekaary level provide karne aur banaye rakhne ki Ability hai. Networking me, ye ek network ki viphalata se ubarane ki Ability hai. Decision Support System (DSS) ek information system hai jo vyavasaay ya sangathanaatmak nirnay lene aur yojana (planning) Gatividhiyon ka sammeaningn karti hai. [Q345–Q348]

- Ek Analog computer hala ki go rahi samasy ko model karne ke liye nirantar physical ghatanaon, jaise electrical, mechanical ya haaidrolik amount ka use karta hai. ek conventional kaara speedometer ek Analog computer ka example hai. computer software data ya computer Instructionson ka ek sanplanet hai jo computer ko bataata hai ki kaise kaam karna hai. computer programme ko saamohik roop se computer Software kaha jata hai. computer ke main component Input Unit, Output Unit, Memory Unit aur Central Processing Unit hain. computer aarkitekchar me 'Basic Unit' koi recognition mila component nahi hai. [Q349–Q352]

11. IT Companies, Social Media, E-Commerce & Cloud Computing

- Microsoft Corporation (Microsoft) ek bahuraashTreeya technology Company hai jo computer software (jaise vindow oesa, emaesa ophisa soota) developed karti hai. Azim Premji WIPRO (Wipro) limited ke sansthapak aur long time tak adhyaksh rahe hain. (unke son rishad premajee ne baad me kaaryakaaree adhyaksh ke roop me padabhaar sanbhaalaa). Google ke seeeeo sundar pichaaee hain (A-2). Twitter (Twitter) ka development maarch 2006 me Jack Dorsey, nooha glaasa, bija stona aur ivaana viliyams dwara kiya gaya tha. [Q353–Q356]
- Facebook (aba metaa pletaphorma, inka. Ka hissaa) ki saha-sthapanaa 2004 me Mark Zuckerberg ne ki thi. LinkedIn ek vyavasaay aur rojagaara-kendrita onalaain peshevar mancha hai jo peshevaron ke liye ek Social Networking Site ki tarah kaam karta hai. E-Commerce vyavasaayon ko din ke 24 ghante, saptaah ke 7 Days khule rahane ki allow karta hai, jisse ye nishchit parichaalan ganton wale physical stonon ke comparison me highly laabhprad ho jata hai. Ema-jankshana (M-Junction - SAIL aur Tata Steel ek joint udyama) ek bade B2B onalaain maarketaples ki tarah kaam karta hai aur lenaden moolya ke hisaab se India ki sabse badi e-Commerce Company hone ka daavaa karta hai. [Q357–Q360]
- Flipkart, aleebaaba aur amejen sabhi vyavasaaya-se-upabhoktaa (B2C) e-Commerce vyavasaayon ke example hain. Ve mainly directly vyaktigat consumers ko utpaad bechate hain. MP Online Madhya Pradesh government ka ek morerit Portal hai jiska purpose G2C (government se naagarika) aur B2C seavaan provide karna hai, ye specially ee-prokyoramenta Portal nahi hai. Cloud computing seavaon ko conventional roop se teen main shreniyon me vibhaajit kiya gaya hai: inphraastrakchar eja e sarvis (IaaS), pletaphorm eja e sarvis (PaaS), aur Software eja e sarvis (SaaS). SaaS application Internet par vitarit kiye jate hain aur inhen mobile phone sahita kisi bhi kanekted divaais se eksesa kiya ja sakta hai. (statement 2 sahi hai). [Q361–Q364]
- Skaaedraaiv (SkyDrive) Microsoft ki vyaktigat cloud storej service ka former naam tha. Tredamaark Contention ki wajah se, Microsoft ne 2014 me official taur par iska naam badalakar "vanadraaiva" (OneDrive) kar diya. Cortesa ek opana-sorsa lo-Codes pletaphorm hai, eaaee varchual asistent nahi. 'projekta loona' (Project Loon) graameen aur dooradaraaj ke kshetron me vaayarales Internet ka use provide karne ke purpose se Google X dwara ek research pahala thi. DACNET e-Agriculture se related hai. Ye India me krishi, sahakaarita aur kisaan kalyan department ka E-Governance web Portal hai. [Q365–Q368]

⚠ EXAM TRAP — Q354–Q367 — Company/Service Facts Can Change

- ▶ CEO/company-head, biggest e-commerce company, cloud-service branding (SkyDrive→OneDrive) aur discontinued projects (e.g., Project Loon) time-sensitive/source-period facts hain.
- ▶ Q361 me Flipkart, Alibaba aur Amazon ko source B2C/e-commerce context me multiple correct options ke saath treat karta hai.

12. Web Acronyms, Logic Gates & Emerging Technologies

- Aaebeem (IBM) ka full form International Business Machine korporeshan hai. Ye ek American bahuraashTreeya technology aur paraamarsh Company hai. HTTP ka full form Hypertext Transfer Protocol hai. Ye World Wide Web par information systems ke liye use kiya jane wala basic application Protocols hai. World Wide Web par data tak reach aur sthanaantarit karne ke liye HTTP (Hypertext Transfer Protocol) primary Protocols hai. SMTP ka use e-mail ke liye kiya jata hai. HTML ka full form Hyper Text Markup Language hai. Ye web browser me display hone wale web pejon ko banane aur design karne ke liye use ki jane wali standard language hai. [Q369–Q372]
- XML ka matalab Extensible Markup Language (eXtensible Markup Language) hai. Ye usually data storage aur sancharan ke liye use kiya jata hai. URL ka full form Uniform Resource Locator hai. Ye World Wide Web par Documentson aur resources ka detect hai. FTP ka meaning File Transfer Protocol hai. Iska use network par computer fileon ke hastaantaran ke liye kiya jata hai. SMPS ka full form Switchda-MODE paavar saplaaee hai. Ye computer ko essential kam Voltseja deesee (DC) provide karne ke liye electrical Shakti ko kushalataapoovrak convert karta hai. [Q373–Q376]
- UPS ka full form 'Uninterrupted Power Supply' hai. Ye ek device hai jo main bijalee viphal hone par baitaree dwara aapaatakaaleen Shakti provide karta hai. OR Gate ek digital logic sarkit hai jiska output HIGH (1) ho jata hai yadi iska ek ya zyada input HIGH (1) ho. Official pareeksha ke nihita aarekh ke basis par: A aur 1 ka output = 1 (Q1), B aur 1 ka output = 1 (Q2). Q1 aur Q2 ka output = C = 1. Kevala NAND geta use karke ek Exclusive-OR (XOR) geta banane ke liye, kam se kam 4 NAND geta ki requirement hoti hai. [Q377–Q380]
- World bhara me kam service wale samudaayon me digital saaksharata ke baare me jaagarookata paidaa karne aur use badhaava dene ke liye hara saala 2nd December ko world computer saaksharata divasa manaay jata hai. computer saaksharata divasa 2nd December ko manaay jata hai. Aakash shiksha ki propertyvatta badhane ke liye India government dwara khareede gaye kam laagat wale Tableta Personal computer ka naam hai, jise 2011 me launch kiya gaya tha. development ke liye information technology ke use ko badhaana Andhra Pradesh Government ki rananeeti thi. Ye chandabaaboo naayadoo ke mukhyamantree kaala (1999-2004) ke under start kiya gaya tha. [Q381–Q384]
- Information technology word data adhiplanetna, information, security etc. ke liye network kanekting hardware (Connectivity + hardware) ke nishpaadan se related hai. Soshal meediy optimaaijeshan (SMO) kisi organisation ke sandesh aur onalaain upasthiti ko prabandhit karne aur badhane ke liye soshal meediy saaiton ka use hai. Iska main point aasaanee se prachaar (publicity) banaana hai. Collaborative Filtering (Collaborative filtering) ek anushansa system hai jo users ke

pichhale behavior aur saamohik praathamikataaon ke basis par bhavishyavaanee karti hai. Iska widely amejena, yootyoob aur netaphliks jaise pletaphaarmon dwara use kiya jata hai. Smaart vocha jaise pahanane yogya device jeepeees (GPS) ko traika karke kisi vyakti ke sthana ki pahachaan kar sakte hain. Ve vyakti ki neenda ke paitarn ki bhi nigaraanee kar sakte hain. Shravan device (hearing aids) shravanabaadhiton ki sahaayata ke liye klaasik pahanane yogya device hain. Isliye teenon work sahi hain. [Q385–Q388]

- Ye parivisible Internet par communication aur baatacheet karne wale sensara, Software aur network kanektivitee ke saath embeded physical objects (devices, ghareloo devices, kaaron) ka description karta hai. Ye Internet of Things (IoT) ki accurate paribhaasha hai. 5G ko iski fast Gati, kam vilanbata (latency) aur ek hi time me kai devices ko jodne ki Ability se pahachaana jata hai. Di gayi concept ka description karne ke liye sabse upayukt word 'Metaverse' hai. Metaverse Internet ka ek bhavishy ka punaraavritti hai jisme lagaataara, imarsiv aabhaasee world shamil hai jahaan People aabhaasee objects par sanpatti ke moreaar ka prayog kar sakte hain. Vidya vaahinee project me country bhara ke primary government skoolon me computer Education (IT Education) le jana shamil hai. [Q389–Q392]

✓ Source coverage: Q1–Q392